

Notes: Hopenhayn (Econometrica, 1992)

Entry, Exit, and Firm Dynamics in Long Run Equilibrium

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1 Big picture

- **Question.** How can a competitive industry have persistent entry, exit, firm growth, firm contraction, and cross-sectional heterogeneity in size and value even when aggregate conditions are stationary?
- **Core idea.** Firms face idiosyncratic productivity shocks. After observing their own productivity state, incumbent firms decide whether to continue or exit. New firms can enter by paying a sunk entry cost and drawing an initial productivity state. In long-run equilibrium, entry replaces exit, so aggregate industry variables are stationary even though individual firms keep moving.
- **Main contribution.** The paper builds a tractable stationary competitive equilibrium model of industry dynamics. It extends long-run competitive industry equilibrium from a representative-firm zero-profit condition to a framework with endogenous entry, endogenous exit, heterogeneous firms, and a stationary distribution of firm states.
- **Key mechanism.** Selection operates through the exit rule. Low-productivity firms exit; high-productivity firms survive. The invariant distribution of firms is therefore not the same as the distribution of entrants. It reflects the interaction between entry, exit, and the Markov process governing productivity.
- **Why it matters.** The model gives an equilibrium foundation for empirical regularities in plant and job turnover: large flows of firms/jobs are created and destroyed even when aggregate employment and output are stable.
- **Economic takeaway.** Long-run industry equilibrium is not a static object. A stationary industry can contain continuous micro-level reallocation, and industry primitives - demand, entry costs, fixed costs, productivity persistence, and shock distributions - determine turnover, size distributions, profits, and firm values.

2 Incremental contribution of the paper

- **From static long-run equilibrium to dynamic stationary equilibrium.** Traditional competitive equilibrium uses entry until expected profits are zero, but usually abstracts from firm turnover and heterogeneity. Hopenhayn keeps the zero-profit logic but embeds it in a dynamic stochastic environment.
- **Endogenous entry and exit in one equilibrium object.** The model jointly determines the exit cutoff, the mass of entrants, prices, wages, aggregate output, aggregate input demand, and the invariant distribution of firms.
- **Selection-based firm dynamics.** The paper shows how firm-level productivity shocks generate survival, growth, and exit patterns through optimal stopping. Exit is not imposed mechanically; it is the solution to the incumbent firm's dynamic problem.
- **A tractable benchmark for later firm-dynamics work.** The model becomes a canonical workhorse for heterogeneous-firm industry equilibrium, later extended to investment, labor adjustment costs, firing costs, productivity dynamics, and policy evaluation.
- **Comparative statics of turnover.** The paper studies how changes in primitives shift the stationary distribution and turnover rate. This turns the model from a pure existence result

into a tool for thinking about industry differences in entry and exit.

- **A bridge between empirical facts and equilibrium theory.** The paper is motivated by evidence that firm-specific uncertainty dominates firm-level dynamics and that entry and exit rates vary systematically across industries. It provides an equilibrium explanation for these facts.

3 Empirical motivation and facts

The paper begins from empirical facts on U.S. manufacturing turnover. Roughly a third of jobs and more than forty percent of firms disappear over five-year periods and are replaced by new jobs and new firms. The model is designed to explain such high reallocation without relying on aggregate shocks.

3.1 Two motivating observations

1. **Firm-specific uncertainty dominates firm dynamics.** Individual firms grow, shrink, survive, or exit mainly because of idiosyncratic shocks rather than common aggregate shocks.
2. **Entry and exit rates are correlated across industries.** Industries with high entry also tend to have high exit. This suggests that industry primitives can generate high-turnover and low-turnover steady states.

3.2 Why a stationary model is useful

A stationary model can describe a long-run industry where aggregate quantities are constant, but micro-level reallocation is ongoing. This is important because aggregate stability does not imply micro stability. In Hopenhayn's model, aggregate output, input use, price, wage, and the distribution of firm states can be constant, while individual firms continually enter, exit, grow, and contract.

4 Relation to the literature

4.1 Earlier stochastic size-distribution models

Earlier work by Hart and Prais, Simon and Bonini, and Adelman showed that stochastic firm-size processes can match observed size distributions. Those models were largely statistical rather than equilibrium models: they specified firm-size dynamics directly rather than deriving them from firms' optimization and market clearing.

4.2 Competitive industry dynamics

Lucas and Prescott introduced competitive equilibrium industry dynamics, but without the same firm-level heterogeneity and endogenous turnover emphasized here. Brock and Vernon Smith developed dynamic entry-exit models, but firms were identical in size and entry/exit disappeared in the limit. Hopenhayn's contribution is to place heterogeneous firms and persistent turnover inside a competitive equilibrium framework.

4.3 Connection to Jovanovic's learning and selection model

Jovanovic's model is the most direct predecessor. In Jovanovic, firms learn about their efficiency and exit when expected value is low. Hopenhayn differs by embedding a similar selection logic into a stationary competitive industry equilibrium with entry, exit, and market clearing. The paper therefore converts a theory of firm selection into an industry-equilibrium model.

5 Model environment

5.1 Timing and agents

The industry is populated by a continuum of firms. There is no aggregate uncertainty, but firms face idiosyncratic productivity shocks. Time is discrete and infinite. Each period:

1. incumbents observe their productivity state;
2. incumbents decide whether to continue or exit;
3. continuing firms choose output and inputs given prices;
4. potential entrants decide whether to pay an entry cost;
5. entrants draw initial productivity states from an entrant distribution;
6. productivity evolves according to a Markov transition function.

5.2 Productivity state

Let $\varphi \in S$ denote a firm's idiosyncratic productivity state. In the paper, $S = [0, 1]$. Higher φ means a more productive or more profitable firm. The productivity state evolves according to a transition function

$$F(d\varphi' \mid \varphi),$$

which gives the distribution of next period's productivity conditional on current productivity.

5.3 Demand and input supply

The industry faces inverse demand and input supply functions. Let Q denote aggregate output and N aggregate input use. Then

$$p = D(Q), \quad w = W(N),$$

where p is the output price and w is the input price. Demand is decreasing in aggregate output, and input supply is increasing in aggregate input demand. This embeds the industry in a partial-equilibrium market environment.

5.4 Production and profits

For a firm with state φ , output and input demand are chosen optimally given prices. The firm's static profit is

$$\pi(\varphi, p, w) = \max_{q, n} \{pq - wn : q \leq f(n, \varphi)\}.$$

The important monotonicity property is that static profits are increasing in productivity φ , increasing in output price p , and decreasing in input price w .

5.5 Entry cost and entrant distribution

Potential entrants pay a sunk cost c_e to enter. After entry, they draw an initial state from a distribution ν . Since the entry cost is sunk, it affects entry but not continuation decisions after entry.

6 The incumbent firm's dynamic problem

6.1 Value function

Given prices (p, w) , an incumbent with state φ chooses whether to exit or continue. If it exits, its continuation value is zero. If it continues, it earns current profit and receives the discounted expected value of future continuation. The value function is

$$v(\varphi; p, w) = \pi(\varphi, p, w) + \max \left\{ 0, \beta \int v(\varphi'; p, w) F(d\varphi' | \varphi) \right\}.$$

In the stationary equilibrium section, prices are pinned down by the invariant measure of firms, so the paper writes $v(\varphi, \mu)$, where μ is the distribution of incumbent firms.

6.2 Exit rule

A firm exits when the continuation value is negative or zero. The exit condition can be written as

$$\int v(\varphi'; \mu) F(d\varphi' | x) = 0,$$

where x is the marginal state. Firms below the cutoff exit, and firms above the cutoff continue. The cutoff exists because firm value is increasing in productivity. Economically, low-productivity firms have insufficient expected future value to justify staying.

6.3 Selection intuition

The exit rule creates selection. Among entrants, many low-productivity firms may enter, but low-productivity firms are more likely to exit. Therefore older cohorts have better selected productivity distributions than younger cohorts. This is one reason the model can generate age-dependent survival and size patterns.

7 Industry equilibrium with a given initial distribution

7.1 Recursive equilibrium

For a given initial measure of firms, an equilibrium consists of prices, firm policies, entry, exit, and a law of motion for the distribution such that:

1. firms solve their static production and dynamic exit problems;
2. entrants enter only when expected entry value equals or exceeds the entry cost;
3. goods and input markets clear;
4. the distribution of firms evolves consistently with incumbent survival and entrant inflows.

7.2 Theorem 1: existence and uniqueness along the transition

The paper proves that, given any initial distribution, there exists a unique equilibrium path. This result matters because the stationary equilibrium is not just a formal fixed point; it is connected to a well-defined dynamic equilibrium.

7.3 Law of motion for firms

Given an exit rule, the distribution next period combines surviving incumbents and new entrants. Conceptually,

$$\mu_{t+1}(A) = \int_{\text{survivors}} F(A \mid \varphi) \mu_t(d\varphi) + M_t \nu(A),$$

where M_t is the mass of entrants and ν is the entrant distribution. This law of motion is the key accounting identity behind stationarity.

8 Stationary competitive equilibrium

8.1 Definition

A stationary competitive equilibrium is a tuple

$$(x^*, M^*, \mu^*, p^*, w^*)$$

such that:

1. x^* is the optimal exit cutoff;
2. M^* is the mass of entrants;
3. μ^* is invariant under the law of motion generated by x^* and M^* ;
4. prices (p^*, w^*) clear output and input markets given μ^* ;
5. the expected value of entry equals the sunk entry cost when entry is positive.

8.2 Invariant measure

For a fixed exit cutoff x and mass of entrants M , the invariant distribution satisfies

$$\mu = \hat{P}_x \mu + M \nu,$$

where $\hat{P}_x$ is the transition operator for incumbent firms that survive under cutoff x . The invariant measure summarizes the entire cross-sectional distribution of incumbent firms.

8.3 Entry condition

With positive entry, expected entrant value equals the entry cost:

$$\int v(\varphi, \mu) \nu(d\varphi) = c_e.$$

If expected entry value is below entry cost, there is no entry. If it is above entry cost, entrants flow in until market-clearing price and wage effects reduce entry value back to c_e .

8.4 Exit condition

The marginal firm is indifferent between continuation and exit:

$$\int v(\varphi', \mu) F(d\varphi' | x) = 0.$$

This condition pins down the cutoff for a given invariant measure. Because the value function is increasing in productivity, the exit policy is a cutoff rule.

9 Existence logic for stationary equilibrium

The paper's existence proof is constructive and uses two curves, $M_1(x)$ and $M_2(x)$.

9.1 $M_1(x)$: entrants needed for the exit rule to be optimal

For each possible exit cutoff x , $M_1(x)$ is the mass of entrants such that the value of the marginal firm at x is exactly zero. Intuitively, changing the mass of entrants changes aggregate output and input demand, which changes prices and wages, which changes incumbent values. Thus $M_1(x)$ describes the entry mass that makes x an optimal exit cutoff.

9.2 $M_2(x)$: entrants needed for free entry

For each exit cutoff x , $M_2(x)$ is the mass of entrants such that expected entrant value equals the entry cost:

$$\int v(\varphi, m(x, M_2(x))) \nu(d\varphi) = c_e.$$

Thus $M_2(x)$ describes the entry mass that satisfies free entry for a given exit rule.

9.3 Equilibrium as intersection

A stationary equilibrium with positive entry exists when

$$M_1(x^*) = M_2(x^*)$$

for some cutoff x^* . At that point, the same mass of entrants simultaneously makes the exit rule optimal and makes entry value equal to entry cost.

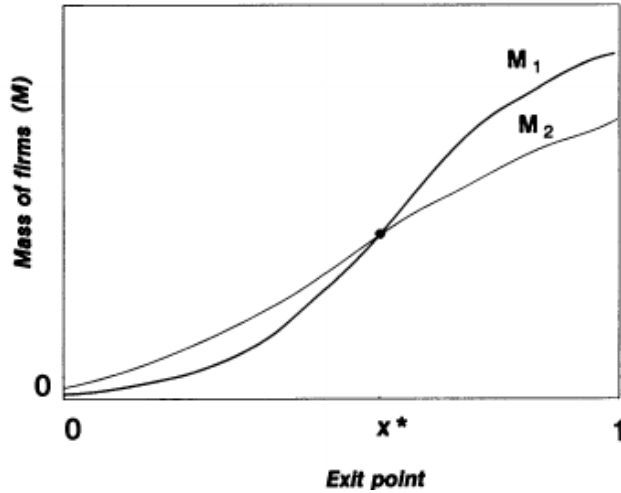


Figure 1: Existence of stationary equilibrium. The stationary equilibrium is represented by the intersection of M_1 and M_2 . M_1 gives the entry mass consistent with an optimal exit cutoff, while M_2 gives the entry mass consistent with free entry.

9.4 Theorem 2: existence of stationary competitive equilibrium

Theorem 2 proves that there exists a stationary competitive equilibrium for the industry. The proof handles two cases. If the M_1 and M_2 curves intersect, the equilibrium has positive entry and exit. If no positive-entry intersection exists, the model can still have an equilibrium with no entry. This is why the theorem establishes existence generally, not only for high-turnover industries.

9.5 Theorem 3: positive entry and exit under technological and demand conditions

Theorem 3 gives conditions under which the stationary equilibrium has positive entry and exit. Economically, the paper wants a nondegenerate equilibrium where firms are continually born and die, rather than a corner case with no entry or no exit.

10 Uniqueness and long-run convergence

10.1 Why uniqueness is nontrivial

Stationary equilibrium is a fixed point involving a distribution, prices, firm values, exit rules, and entry. Multiple stationary equilibria could in principle arise because entry affects prices, prices affect value, value affects exit, and exit affects the distribution.

10.2 Uniqueness conditions

The paper provides two alternative sufficient conditions, labeled U.1 and U.2. These conditions restrict primitives so that the relevant fixed-point mapping is well behaved. Under these conditions, the stationary equilibrium is unique.

10.3 Theorem 4

Theorem 4 states that if either uniqueness condition is satisfied and if an equilibrium with entry and exit exists, then that equilibrium is unique. The result is important for comparative statics: if the stationary equilibrium is unique, changes in primitives have an unambiguous effect on the equilibrium object.

10.4 Ergodic interpretation

Under recurrence and monotonicity assumptions, the distribution of firms converges to a stationary distribution. The stationary distribution is not an arbitrary assumption; it is the long-run outcome of entry, exit, and productivity transitions.

11 Cross-sectional implications

11.1 Size distribution

Firm size is increasing in productivity because higher-productivity firms choose higher output and input demand. Since the stationary distribution of productivity is shaped by selection, the model implies a stationary firm-size distribution. The size distribution depends on:

- the entrant productivity distribution;
- persistence in productivity shocks;
- the exit cutoff;
- market demand and input supply;
- entry costs and fixed production costs.

11.2 Age distribution

Older firms are selected survivors. Conditional on survival, older cohorts tend to have higher productivity because low-productivity entrants exit early. The model therefore naturally implies differences between young and old firms in survival and size.

11.3 Growth rates

Growth is driven by productivity transitions. A firm that receives a favorable shock expands; a firm that receives an unfavorable shock contracts or exits. The model can therefore generate job creation and destruction within the same stationary industry.

11.4 Entry and exit correlation

In a stationary equilibrium, the mass of entering firms equals the mass of exiting firms after accounting for firm flows. High-turnover industries have both high entry and high exit, matching the empirical observation that entry and exit rates are positively correlated across industries.

12 Profits and firm value

12.1 Profit distribution

Because productivity differs across firms, profits differ across firms. The stationary distribution of productivity generates a stationary distribution of profits. High-productivity firms earn higher profits; low-productivity firms earn low or negative continuation value and exit.

12.2 Firm value distribution

Firm value is increasing in productivity. Therefore the model implies a cross-sectional distribution of firm values. The paper emphasizes that the distribution of values is not just a transformation of static profits; it includes continuation options and the value of future productivity shocks.

12.3 Proposition 7: average profits of incumbents

The paper shows that the average profits of incumbent firms exceed the average profits of entering firms net of entry cost in a way linked to selection. Incumbents are a selected group, so their expected values and profits differ from entrants.

12.4 Proposition 8: average value of firms

The average value of incumbent firms exceeds the capitalized value of entry costs under the model's assumptions. The economic reason is again selection: entrants are ex ante draws, while incumbents are firms that survived the selection process.

13 Comparative statics

13.1 Technological improvements

A productivity-improving change increases firm values and tends to affect both the exit cutoff and entry. The exact direction of turnover can depend on how the shock changes the relative value of incumbency and entry. A general lesson is that productivity improvements can alter not only output and prices but also the composition and turnover of firms.

13.2 Demand shifts

An outward demand shift raises output price for a given distribution, increasing profitability and entry incentives. More entry changes the invariant distribution and can increase competition. The final stationary effect combines direct profitability effects and general equilibrium price effects.

13.3 Input supply shifts

A higher input cost lowers profits and continuation values. This can raise exit, reduce entry, and alter the stationary size distribution. Since firms differ in productivity, cost changes also change selection: marginal firms are the most affected.

13.4 Entry costs

A higher sunk entry cost lowers entry incentives. Under conditions in the paper, an increase in the fixed cost of entry shifts the stationary equilibrium by reducing entry and changing the equilibrium distribution. Higher entry barriers can reduce turnover and make the industry more concentrated among incumbents.

13.5 Fixed operating costs

Higher fixed operating costs reduce continuation values, especially for low-productivity firms. This can raise the exit cutoff, causing more selection among incumbents. However, general equilibrium effects through entry and prices matter for the final stationary distribution.

14 Detailed economic intuition of the model

14.1 Why low-productivity firms exit

The firm compares the value of staying with the outside option of exit. Because staying preserves the option to receive future productivity improvements, some firms with low current profits may continue. But if current productivity is too low, the option value is insufficient and exit is optimal.

14.2 Why entry occurs despite exit risk

Entrants pay a sunk cost to receive a productivity draw and access to future profits. Even though many entrants will exit, entry can be profitable ex ante because some entrants receive good productivity draws and become high-value incumbents.

14.3 Why the stationary distribution differs from the entrant distribution

The stationary distribution over incumbent firms overweights successful survivors and underweights failed entrants. It is generated by repeated filtering: entrants draw from ν , low types exit, and surviving types transition according to F .

14.4 Why the model has simultaneous creation and destruction

In the steady state, every exiting firm is replaced by entrants in aggregate, but firm identities change. Thus net aggregate employment can be constant while gross job creation and destruction are large.

15 Identification of primitives from data: what the model suggests

The paper is theoretical, but it points to empirical objects that later work can estimate or discipline:

- the productivity process $F(\varphi' \mid \varphi)$;
- the entrant productivity distribution ν ;
- fixed operating costs and sunk entry costs;
- demand elasticity and input supply elasticity;
- the mapping from productivity to employment and output;

- survival and exit rates by age and size;
- the stationary size and age distributions.

16 Comparison with Jovanovic and Ericson-Pakes

16.1 Compared with Jovanovic

Jovanovic emphasizes learning about firm efficiency and selection through exit. Hopenhayn shares the selection logic but does not center the model on learning. Instead, productivity evolves according to a Markov process, and the central innovation is the stationary competitive equilibrium with entry and exit.

16.2 Compared with Ericson and Pakes

Ericson and Pakes allow firms to invest in improving future productivity. Hopenhayn's baseline abstracts from investment, so the only dynamic decision is exit. This makes the model analytically tractable. Hopenhayn explicitly notes that investment can be added by expanding the state vector or allowing firm actions to affect the transition function.

17 Policy relevance

17.1 Labor firing costs

The model provides a natural framework for studying policies that affect exit, job destruction, and reallocation. Firing costs, for example, can reduce job turnover, distort exit decisions, lower productivity, and affect welfare. Hopenhayn and Rogerson later use a related framework to study firing costs.

17.2 Entry regulation

Entry costs affect the mass of entrants and the selection process. Policies that raise entry costs may reduce turnover and experimentation with new firms, while policies that lower entry barriers can increase entry and exit simultaneously.

17.3 Misallocation and productivity

Because selection reallocates resources from low-productivity firms to high-productivity firms, policies that impede exit or entry can reduce aggregate productivity. The model is therefore an early theoretical foundation for later misallocation and reallocation research.

18 Limitations

- **No endogenous investment in the baseline.** Firms cannot invest to improve productivity; they only decide whether to stay or exit.
- **No aggregate shocks.** The model focuses on stationary equilibrium without aggregate uncertainty.

- **Single productivity state.** Real firms differ in many dimensions: capital, labor composition, demand, product quality, financial constraints, and management. The baseline collapses heterogeneity into one state.
- **No strategic interaction.** Firms are competitive and atomistic. The model is less suited for oligopolistic industries where individual firms affect prices.
- **No financing constraints.** Entry costs are paid by potential entrants, but the model does not study credit frictions or borrowing constraints.
- **No labor adjustment costs in the baseline.** Employment responds freely to productivity and prices, so the model abstracts from sluggish adjustment.

19 Important equations and how to read them

19.1 Firm value equation

$$v(\varphi, \mu) = \tilde{\pi}(\varphi, \mu) + \max \left\{ 0, \beta \int v(\varphi', \mu) F(d\varphi' | \varphi) \right\}.$$

Meaning: current profit plus the option value of continuation. The max operator captures exit.

19.2 Exit cutoff equation

$$\int v(\varphi', \mu) F(d\varphi' | x) = 0.$$

Meaning: the marginal surviving firm is indifferent between staying and exiting.

19.3 Free-entry equation

$$\int v(\varphi, \mu) \nu(d\varphi) = c_e.$$

Meaning: entrants enter until expected entrant value equals the sunk entry cost.

19.4 Invariant distribution equation

$$\mu = \hat{P}_x \mu + M\nu.$$

Meaning: next period's incumbent distribution equals surviving incumbents plus new entrants.

20 How to use this paper for research

20.1 As a baseline model

This paper is useful when a research question involves entry, exit, survival, turnover, firm-size distributions, or productivity selection. It provides the cleanest benchmark where heterogeneity and turnover arise from idiosyncratic productivity shocks.

20.2 As a platform for policy analysis

Many policies can be interpreted as changing primitives:

- entry regulation changes c_e ;
- labor-market frictions change operating costs and exit incentives;
- subsidies change profits or entry values;
- demand shocks shift $D(Q)$;
- technology shocks change the production function and productivity process.

20.3 As a bridge to empirical firm dynamics

Empirical work can use the model to interpret facts such as:

- exit rates by age and size;
- growth rates by age and size;
- entry and exit correlations across industries;
- productivity distributions among entrants and incumbents;
- job creation and destruction in steady state.

21 Short summary

- Hopenhayn develops a stationary competitive equilibrium model of entry, exit, and firm dynamics.
- Firms differ by idiosyncratic productivity and decide optimally whether to continue or exit.
- Entrants pay a sunk cost and draw initial productivity from an entrant distribution.
- A stationary equilibrium jointly determines the exit cutoff, entry mass, invariant distribution, output price, input price, and aggregate quantities.
- Entry and exit coexist in steady state; aggregate variables are constant because entry replaces exit.
- The model explains how cross-sectional heterogeneity in size, profits, and firm value can persist in long-run equilibrium.
- Selection is central: low-productivity firms exit, while surviving firms are positively selected.
- The model provides comparative statics for how entry costs, fixed costs, demand, and technology affect turnover and the firm-size distribution.
- The framework became a foundation for later quantitative firm-dynamics and policy-evaluation models.

22 Conclusion

22.1 Core conclusion

Hopenhayn shows that long-run competitive industry equilibrium can be dynamic at the micro level. A stationary industry can have constant aggregate output and employment while individual firms continuously enter, exit, expand, and contract. The key equilibrium object is the invariant distribution of firm states, which is shaped by productivity shocks, selection, and entry.

22.2 Economic intuition

The model's intuition is simple but powerful. Entry creates new draws from the productivity distribution. Exit removes low-value firms. Productivity shocks move firms around the state space. Market clearing determines prices and wages. The stationary equilibrium balances all these forces so that the distribution of firms is constant even though individual firms are not.

22.3 Incremental contribution revisited

The paper's main contribution is not only that it models entry and exit, but that it gives entry and exit an equilibrium stationary structure. This makes it possible to study firm turnover, size distributions, and policy changes in a coherent long-run framework.

22.4 Takeaway

The paper is a canonical reference for thinking about firm dynamics as equilibrium reallocation. It teaches that persistent turnover is not a sign of disequilibrium; it can be the normal long-run outcome of selection under idiosyncratic uncertainty.